Odds & Ends: Statistics for Animal Studies

What you know, what to watch out for, and where to go next

Emile Latour

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Part 1: You Know More Than You Think

Mapping your experiments to your toolkit

Every animal study comes down to a few fundamental questions. You've been answering versions of these all semester.

The key decisions are ones you already know how to make:

The three questions that determine your test

1. **What type of outcome?** Continuous, binary, ordinal?
2. **How many groups?** One, two, or three or more?
3. **Are the observations independent?** Or are they paired / repeated on the same subject?

Your course toolkit, translated to the bench

Study design	Groups	Outcome	Test	Key R function
Compare mean to a known reference value	1	Continuous, ~normal	One-sample t-test	`t.test(y, mu = ref)`
Two groups, one timepoint (~normal)	2	Continuous, ~normal	Two-sample t-test	`t.test(y ~ group)`
Two groups, one timepoint, skewed or small n	2	Continuous, skewed	Wilcoxon rank-sum	`wilcox.test(y ~ group)`
Same animal before and after treatment	1 (paired)	Continuous	Paired t-test	`t.test(pre, post, paired = TRUE)`
Three or more dose groups at one timepoint	3+	Continuous, ~normal	One-way ANOVA	`aov()` + post-hoc
Three or more groups, skewed or small n	3+	Continuous, skewed	Kruskal-Wallis	`kruskal.test()`
Does response rate match a historical benchmark?	1	Binary	One-proportion z-test	`prop.test(x, n, p = ref)`
Does response rate differ between two groups? (large n)	2	Binary	Two-proportion z-test	`prop.test()`
Does response rate differ between two groups? (small n)	2	Binary	Fisher's exact	`fisher.test()`
Same animal, binary outcome before and after (paired binary)	1 (paired)	Binary	McNemar's test	`mcnemar.test()`
Association between two categorical variables	2+	Categorical	Chi-square	`chisq.test()`
Continuous predictor and continuous outcome	—	Continuous	Simple linear regression	`lm()` + `tidy()`

The conversation this prepares you for

When you sit down with a statistician, the most useful thing you can do is walk in with clear answers to:

What a statistician wants to know

- What is your **outcome**, and how is it measured?
- What is your **primary comparison** — treatment vs. control? Before vs. after?
- How many **groups**, and are they **independent or paired**?
- What is your **sample size**, and how was it chosen?
- Are there **other variables** that might confound the relationship?

What this course gave you

You now have the vocabulary to answer all of these questions — and to understand the statistician's answer.

You don't need to know the method. You need to know your data and your question.

That's what brings to a productive collaboration.

Part 2: Experimental Design Matters

The decision that can't be fixed in analysis

Imagine you're preparing slides from tumor tissue.

It's a long protocol. You decide to process all **control samples on Monday** and all **treated samples on Wednesday**.

Seems fine — same technician, same reagents, same protocol.

But what if:

- The antibody lot was slightly different?
- Room temperature differed?
- You were more fatigued by slide 30?

The problem: confounding by batch

Now **treatment** and **processing day** are perfectly confounded.

Any difference you observe between groups could be due to:

- The treatment ✓
- The day of processing ✕
- Some combination of both ✕

You cannot separate them in analysis. The damage is done at the design stage.

What is a batch effect?

A **batch effect** is any source of technical variability introduced by when or how samples are processed — not by your biological treatment.

Common sources of batch effects

- Day of experiment / run
- Reagent lot or antibody batch
- Technician (inter-operator variability)
- Equipment calibration drift
- Passage number in cell lines
- Time in culture before harvest

Why cell lines deserve special mention

Cell lines can drift over passages. A control run at passage 12 and a treatment run at passage 20 may not be biologically comparable — even if the protocol is identical.

This is a batch effect with biological consequences.

The fix: blocking

Randomize your treatment and control samples across batches — process some controls and some treated samples together in each run. Now batch effects cancel out across groups rather than accumulating in one of them.

Blocking: turning nuisance into noise

Without blocking: treatment is confounded with day

```
Monday:    Control  Control  Control
Wednesday: Treated  Treated  Treated
```

You can't know if differences come from treatment or day.

With blocking: treatment is balanced across days

```
Monday:    Control  Treated  Control  Treated
Wednesday: Treated  Control  Treated  Control
```

Now any day effect hits both groups equally — it's **noise**, not **bias**.

Note

Blocking is the experimental design analog of the **paired t-test** — you're using structure in the design to reduce unwanted variability.

The principle

Design your experiment so that any batch effect is balanced across your groups.

This is why randomizing *within* batches — not just randomizing overall — matters. Convenience-driven sequencing (all controls first, then all treated) is one of the most common and consequential design errors in bench science.

Part 3: The Replication Problem

Not all replicates are created equal

Biological replicates

Separate biological units — different animals, different cell culture preparations, different patients.

Each is an independent sample from the population of interest.

This is what your sample size n should count.

Technical replicates

Repeated measurements of the same biological unit — running the same sample through the assay three times, imaging the same tissue section twice.

These measure **instrument precision**, not biological variability.

Why it matters statistically

If you treat technical replicates as biological replicates, you **artificially inflate your sample size**.

Example: 3 mice, each measured 5 times.

- True biological n = **3**
- If you treat all measurements as independent: n = **15**

With $n = 15$ your standard errors are artificially small, your t-statistics are inflated, and your p-values are too small.

You will find "significant" results that aren't real.

What the field actually does — and why

An honest conversation

Using technical replicates as if they were biological replicates is **not statistically correct**.

It is, however, **common in practice** — particularly in early-stage, exploratory bench work.

Why? Because biological replicates are expensive, time-consuming, and sometimes require institutional ethics approval for additional animals.

The pragmatic approach many labs take: average technical replicates within each biological unit first, then analyze those averages. This at least preserves the correct n.

What you should do

1. **Be clear** about what type of replicates you have
2. **Report** both biological and technical n in your methods
3. **Analyze** at the level of the biological replicate
4. **Acknowledge** limitations if technical replicates were used due to feasibility constraints

When in doubt — **ask a statistician before you collect data**, not after.

Part 4: Where Your Toolkit Runs Out

Longitudinal animal studies: the naive approach fails

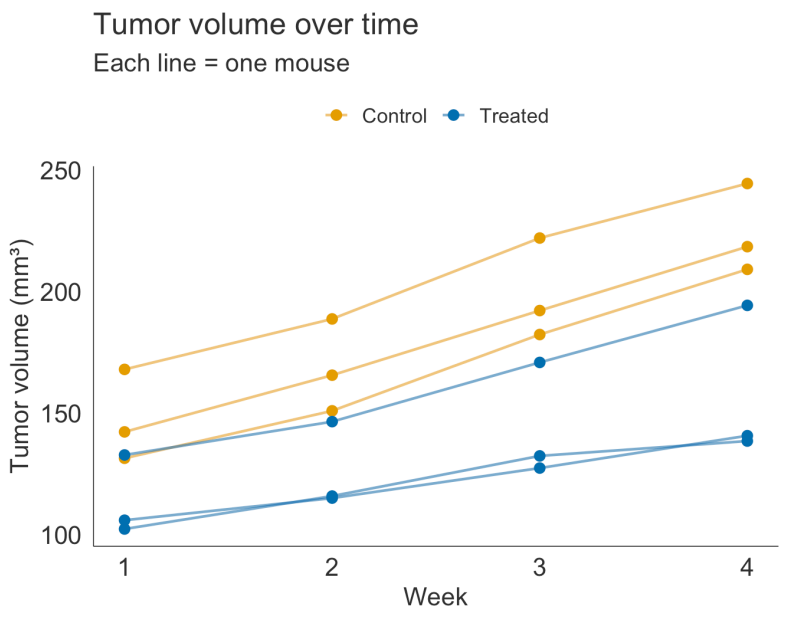
Many animal studies measure the **same outcome repeatedly over time** — tumor volume weekly, body weight daily, behavioral scores at multiple timepoints.

The tempting but wrong approach

Run a t-test at **each timepoint separately**.

Two problems:

1. **Multiple comparisons** — testing at 4 timepoints with $\alpha = 0.05$ means your family-wise false positive rate is already $> 18\%$ under the null, just like the ANOVA post-hoc problem
2. **Non-independence** — measurements from the same mouse at week 1 and week 2 are **correlated**. A mouse with high tumor volume at week 1 will likely have high tumor volume at week 2. Standard errors that assume independence are **too small**.



The solution: linear mixed models

Linear mixed models (LMM) handle both problems at once. The key idea: give each animal **its own baseline**.

The model has two kinds of effects:

Fixed effects

The parameters we care about — treatment group, time, their interaction. These are estimated the same way as in regular regression.

Random effects

Animal-specific offsets from the overall intercept. Each mouse gets its own "starting point," which absorbs the within-animal correlation.

```r
library(lme4)

model_lmm <- lmer(
  volume ~ week * treatment + (1 | mouse_id),
  data = tumor_data
)

summary(model_lmm)
```

Tip

The `(1 | mouse_id)` term is the random intercept — it tells R that observations from the same mouse are correlated.

You don't need to know how to fit this yet. You need to know **when to ask for it**.

When to ask for a mixed model

The signal that you need more than a t-test or ANOVA

If the same subject appears more than once in your dataset, you probably need a mixed model.

Design	Repeated measures?	Method
Two groups, one sacrifice timepoint	No	t-test / Wilcoxon
Pre/post on same animal (two timepoints)	Yes (2 only)	Paired t-test
Same animal measured weekly for 8 weeks	Yes (many)	Linear mixed model
Cells from same animal run in triplicate	Yes (technical)	Mixed model preferred; averaging within animal then t-test is a common simplification

What to tell a statistician

"I have repeated measurements on the same animals over time. I think I need a mixed model but I'm not sure how to set it up."

That sentence will make any statistician very happy to work with you.

Part 5: Power and the 3Rs

Why sample size matters more in animal studies

You've used `pwr` to calculate sample size for t-tests and ANOVA. The same tools apply to animal studies — but the **stakes are different**.

In a clinical trial, enrolling more patients than needed is a resource question.

In an animal study, it's an **ethical question**.

The 3Rs framework (Russell & Burch, 1959)

- **Replace** — use non-animal alternatives where possible
- **Reduce** — use the minimum number of animals needed to achieve scientific objectives
- **Refine** — minimize suffering and improve animal welfare

Institutional animal ethics review and many journals now require a power justification for n.

The two failure modes

Too few animals (underpowered) Study misses a real effect. Animals were used and the science learned nothing. This violates *Reduce*.

Too many animals (overpowered) More animals than needed to answer the question. Also violates *Reduce*.

Power analysis is not just a statistics formality — **it is an ethical obligation.**

Power analysis with tools you already have

You already know `pwr.t.test()`. The inputs are the same for animal studies:

```r
library(pwr)

# How many mice per group do I need?
# Expecting a large effect (d = 0.8),
# alpha = 0.05, power = 0.80

pwr.t.test(
  d         = 0.8,
  sig.level = 0.05,
  power     = 0.80,
  type      = "two.sample"
)
```

```
     Two-sample t test power calculation

              n = 25.52458
              d = 0.8
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number in *each* group
```

Result

26 mice per group — about 52 total.

This is the minimum to detect a large effect with 80% power. For a medium effect ($d = 0.5$), you would need substantially more.

The hardest part: choosing your effect size

> *"I don't know what effect size to expect. What should I use?"*

Practical approaches

- **Pilot data** from your own lab
- **Published literature** in similar models
- **Smallest effect worth detecting** — what difference would be biologically or clinically meaningful?
- **Cohen's conventions** (d = 0.2 / 0.5 / 0.8) as a last resort — but justify why

What not to do

- Don't run the experiment first and then compute power from your results — **post-hoc power is circular**
- Don't choose n by convention ("we always do 5 per group") without justification

Pilot experiments are valuable precisely because they give you an estimate of effect size and variability before you commit to a full study.

Putting it all together

Before you run your experiment:

1. Define your outcome and primary comparison clearly
2. Identify your design: independent groups? paired? longitudinal?
3. Think about batch effects — plan to randomize within batches
4. Distinguish biological from technical replicates — know your true n
5. Run a power analysis — justify your n for ethics review and grant applications

When the data are in:

6. Apply the test that matches your design (use the table!)
7. If you have repeated measures — ask for help with mixed models
8. Report biological n, test statistic, p-value, and effect size

The most valuable thing this class gave you

You can now walk into a meeting with a statistician and have a real conversation.

You know what your outcome is. You know whether it's paired. You know what a p-value means and what it doesn't. You know that sample size requires justification.

That puts you ahead of most people who show up with their data already collected and a question that can no longer be answered.

On the horizon: two designs worth knowing exist

Two extensions of ANOVA come up often in experimental biology. You don't need to know how to run them today — but you should know when to look them up.

Two-way ANOVA

When you need it: Two categorical explanatory variables, one continuous outcome.

Example: Does tumor volume differ by *treatment* AND by *sex*? And does the treatment effect differ between sexes? That last question is an **interaction**.

R function: `aov(y ~ A * B, data = data)`

Repeated measures ANOVA

When you need it: Same subjects at 3+ timepoints; simpler alternative to a full mixed model when data are complete and balanced.

Example: Body weight at baseline, week 2, week 4, and week 8 in the same animals.

R function: `aov(y ~ time + Error(id/time), data = data)`

Where to learn more

Free online textbook

Zhang & Wang (2017), *Advanced Statistics Using R*, Ch. 7

Clear, R-based, covers one-way, two-way, and repeated measures ANOVA.

advstats.psychstat.org/book/ZhangWang2017.pdf

Penn State open course notes

STAT 502 — Analysis of Variance and Design of Experiments

online.stat.psu.edu/stat502

STAT 503 — Design of Experiments

online.stat.psu.edu/stat503